Question 1. Frequentist vs. Bayesian Perspectives

To evaluate the localized cooling effect of a climate engineering intervention (e.g., stratospheric aerosol injection), scientists compared the mean temperatures of an experimental region and a control region. The latest observational analysis reports an apparent cooling effect of

$$d = 0.6\,^{\circ}\mathrm{C}.$$

Due to instrumental noise, uneven sampling networks, and natural climate variability, the standard error of this estimated difference is $\sigma_d = 0.4\,^{\circ}\mathrm{C}$. Both frequentist and Bayesian statisticians accept these numbers, but they profoundly disagree on what may be said next.

(a) **The Frequentist View**

- Calculate the probability that measurement noise and natural variability alone would produce an apparent cooling gap of $0.6\,^{\circ}\mathrm{C}$ or larger if the intervention were truly ineffective (i.e., true cooling $d_{\text{true}} = 0$). You may express your answer using the standard normal cumulative distribution function $\Phi(z)$.
- Furthermore, explain why a frequentist strictly declines to answer the question, "What is the probability that the intervention is truly effective?" Base your reasoning on the frequentist definition of what kind of quantity is allowed to carry a probability.

(b) **The Bayesian View**

- Assume a flat prior, meaning there is no reason beforehand to favor either a cooling or warming effect. Write down the posterior distribution of the true cooling effect d given the observed data.
- Then, calculate the posterior probability that the intervention is truly effective (true $d > 0$). How do the conditional forms of the probabilities differ between the two camps? Express this distinction mathematically using the form $\Pr(A \mid B)$.

(c) **The Common Pitfall**

- Suppose a science news article reports: *"According to statistical analysis, there is only a 6.7% probability that the climate intervention has no effect."* (Note: $\Phi(-1.5) \approx 0.067$). Explain the fundamental statistical error in this statement.
- Using your understanding from parts (a) and (b), rewrite the conclusion into two statistically rigorous sentences: one strictly following the frequentist logic, and one following the Bayesian logic.

(d) **Informative Prior**

- Suppose another research team warns that, based on past laboratory experiments and small-scale climate models, this specific technology often triggers a weak unexpected "warming" rather than cooling. Consequently, the prior distribution is no longer flat, but is instead a Gaussian distribution centered on a negative value.
- Qualitatively explain how this new prior information would alter the Bayesian posterior probability calculated in (b). Finally, would the frequentist calculation (the p-value) in (a) change due to this new prior information? Briefly explain why or why not.

Question 2. Monte Carlo Derivation of the t-Distribution

In class, we learned that when the sample size N is small, the standardized distribution of the sample mean $\overline{x}$ is no longer a standard normal distribution (z-distribution), but instead follows a t-distribution with heavier tails. The course notes define the t-statistic as:

$$t = \frac{\overline{x} - \mu}{\sqrt{s^2/(N-1)}}$$

where

$$s^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \overline{x})^2$$

is the uncorrected (biased) sample variance.

This question asks you to construct the distribution of this t-statistic yourself using a Monte Carlo simulation drawn from a standard normal population, and to compare it against the theoretical analytical solution provided in the notes.

(a) **Monte Carlo Simulation**
Follow these steps to generate the empirical distribution:

- Assume a standard normal population $\mathcal{N}(\mu = 0, \sigma = 1)$.
- Perform $M = 10,000$ independent Monte Carlo experiments.
- In each experiment, randomly draw a sample of size $N = 5$.
- Calculate the sample mean $\overline{x}$ and the biased sample variance s^2 for the sample.
- Compute the t-statistic for each experiment.
- Plot the $10,000$ resulting t-values as a probability density histogram.

(b) **Analytical Comparison**
Overlay the theoretical probability density function (PDF) of the t-distribution for $\nu = 4$ on your histogram. The theoretical formula given in the notes is:

$$f(t) = f_0(\nu)\left(1 + \frac{t^2}{\nu}\right)^{-\left(\frac{\nu+1}{2}\right)}$$

(Note: You may use built-in functions like `scipy.stats.t.pdf` *in Python or* `tpdf` *in MATLAB to plot this theoretical curve, as they inherently compute and include the normalization constant* $f_0(\nu)$*.)*

(c) **Discussion**
Address the following points based on your results:

- Does your Monte Carlo histogram closely match the theoretical t-distribution curve?
- On the same figure, overlay the PDF of a standard normal distribution $\mathcal{N}(0, 1)$. Compare the "tails" of the two theoretical curves. Briefly explain why a small sample size N causes the t-statistic to exhibit heavier tails, even though the data is drawn from a perfectly normal population. *(Hint: Consider the uncertainty introduced by estimating the denominator* $\sqrt{s^2/(N-1)}$ *from a small sample).*

Question 3. Comparison among Different Numerical Approaches for Linear Regression

Generate $n = 10^6$ synthetic observations from

$$y_i = 2 + 3x_i + \epsilon_i,$$

where x_i is uniformly distributed over $[-1, 1]$, and ϵ_i is normally distributed with mean zero and standard deviation 0.5. Assume that the observations are independent and that ϵ_i is independent of x_i. Use a fixed random seed and report its value.

Fit the model

$$\hat{y}_i = a + bx_i$$

by minimizing

$$J(a, b) = \frac{1}{2n} \sum_{i=1}^{n} (a + bx_i - y_i)^2.$$

1. Implement the following three approaches using the same dataset:

   - Built-in least-squares routine: use `np.linalg.lstsq` with `rcond=None`.
   - Normal equations: construct the design matrix $X = [\mathbf{1}, x]$ and solve

     $$X^\mathsf{T} X \boldsymbol{\beta} = X^\mathsf{T} y,$$

     where $\boldsymbol{\beta} = (a, b)^\mathsf{T}$. Use `np.linalg.solve` rather than explicitly computing the matrix inverse.
   - Batch gradient descent: initialize $a = b = 0$ and update both coefficients simultaneously using the gradient of J evaluated over all observations. Use a learning rate of 0.1, stop when $\|\nabla J\|_2 < 10^{-6}$, and impose a maximum of 10000 iterations.

2. Compare the fitted results.

   For each approach, report the intercept, slope, final objective value, and coefficient differences relative to the built-in least-squares solution. For gradient descent, also report the number of updates performed and whether the stopping criterion was satisfied.

   Plot a subset of 100 observations from the generated dataset together with the three fitted lines. Use the coefficients fitted from all 10^6 observations, and evaluate the lines on an increasing grid of x values over $[-1, 1]$.

   Discuss whether the fitted results agree and explain any differences.

3. Compare computational time.

   Use the same observations and design matrix for all three approaches. Exclude data generation, design-matrix construction, plotting, printing, and the final evaluation of J from the timing.

   For the built-in approach, time the call to `np.linalg.lstsq`. For the normal equations, include the computation of $X^\mathsf{T} X$ and $X^\mathsf{T} y$, and the linear solve. For gradient descent, include coefficient initialization, all gradient evaluations, convergence checks, and updates.

   Perform one untimed warm-up run for each approach, followed by at least five timed runs using `time.perf_counter`. Report the median execution time. Reset the gradient-descent coefficients before every run.

   Compare the execution times and explain the observed differences.